Talkly AI: 5-Day Accelerated Fast-Track Sprint

Decoupling Real Estate, Enabling Multi-Use-Case AI, Self-Serve Trial & Outbound Webhooks

Sprint Duration: 5 Calendar Days | Effective Start: Tomorrow, Saturday, Oct 17 - Wednesday, Oct 21, 2026

1. Executive Objective & Strategic Rationale

TalklyAI currently faces two major strategic constraints: (1) it is technically hardcoded for real estate, and (2) it operates on a high-friction, demo-only enterprise onboarding model with opaque pricing. This 5-Day Accelerated Sprint condenses the original 7-day engineering roadmap into 5 high-impact working days. By Day 5, Talkly will be fully industry-agnostic (Healthcare, EdTech, BFSI, B2B Sales, Real Estate), offer instant self-serve signup with 10 free minutes, and sync live call intelligence directly to any CRM via webhooks.

2. Roadmap Condensation: 7-Day Plan vs. 5-Day Fast-Track

Timeline	Original 7-Day Sprint	Accelerated 5-Day Fast-Track Delivery
Day 1	Multi-Use-Case AI Engine (process_audio.py)	Universal AI Engine + Follow-Up Context Engine
Day 2	Dynamic CSV & Campaign Data Model	Flexible Data Model + Dynamic CSV Custom Fields Ingestion
Day 3	Prompt Studio Industry Templates	1-Click Prompt Templates + Generalized Table UI (Campaigns/Leads)
Day 4	Self-Serve Signup & Free Trial Wallet	Public Self-Serve Registration + Automated 10-Min Free Trial
Day 5	Public Pricing Page	Outbound Webhooks / Zapier Sync + Public Pricing + End-to-End QA
Days 6-7	Outbound Webhooks & Final QA	Merged into Days 4 & 5 for immediate market deployment

3. Detailed Day-by-Day Implementation Breakdown

DAY 1 (Sat, Oct 17): Universal AI Engine & Dynamic Prompt Decoupling

Objective: Decouple real estate from post-call LLM analysis and follow-up email drafting.

Key Implementation Tasks:

- Replace hardcoded ANALYSIS_SYSTEM_PROMPT with universal multi-industry prompt & JSON schema.
- Suppor

Deliverables: Universal speech analytics engine supporting Healthcare, EdTech, BFSI, Sales, and Real Estate.

Files Impacted: backend/process_audio.py, backend/server.py, backend/services/followup_service.py

DAY 2 (Sun, Oct 18): Flexible Data Model & Dynamic CSV/Excel Custom Fields

Objective: Enable tenant companies to upload contacts with arbitrary custom columns and inject them into AI calls.

Key Implementation Tasks:

- Update backend/routes/lead_routes.py to collect all non-standard CSV/Excel columns into custom_fields.
- Update

Deliverables: Dynamic CSV lead uploader and multi-variable Bolna prompt injector.

Files Impacted: backend/routes/lead_routes.py, backend/routes/campaign_routes.py, backend/server.py

3. Detailed Day-by-Day Implementation Breakdown (Cont.)

DAY 3 (Mon, Oct 19): Turnkey Industry Prompt Templates & UI Adaptation

Objective: Give non-technical business owners 1-click prompt scripts and generalize frontend tables.

Key Implementation Tasks:

- Build backend/industry_templates.py with ready-to-use scripts for Healthcare, EdTech, BFSI, Sales, and Real Estate.
- Enhanc

Deliverables:	Prompt Studio Template Library + multi-industry campaign & leads views.
Files Impacted:	backend/industry_templates.py, frontend/app/prompt/page.tsx, campaigns/page.tsx, leads/page.tsx

DAY 4 (Tue, Oct 20): Self-Serve Onboarding & Automated Free Trial Grant

Objective: Remove demo-only friction: allow users to sign up and test call immediately without manual approval.

Key Implementation Tasks:

- Enable instant self-registration in backend/routes/auth_routes.py without pending super-admin gating.
- Automa

Deliverables:	Self-serve signup flow with 10 free minutes automatically credited upon registration.
Files Impacted:	backend/routes/auth_routes.py, frontend/app/register/page.tsx, frontend/app/login/page.tsx

DAY 5 (Wed, Oct 21): Outbound CRM Webhooks, Transparent Pricing & Rollout QA

Objective: Solve the ecosystem integration gap, publish clear pricing, and conduct end-to-end smoke testing.

Key Implementation Tasks:

- Upgrade backend/services/crm_service.py to dispatch real HTTP POST webhooks (Zapier, Make, HubSpot).
- Auto-di

Deliverables:	Production-ready multi-industry AI calling platform with webhooks, pricing, and self-serve onboarding.
Files Impacted:	backend/services/crm_service.py, frontend/app/settings/page.tsx, frontend/app/pricing/page.tsx

4. Sprint Acceptance Criteria & Deliverables Checklist

- [x] Multi-Industry Speech Analytics: Transcript analysis scores any business inquiry (clinic, course, loan, property).
- [x] Flex